

Evaluation Platform — Model Accuracy & Availability Report

Live smoke-test verification of every AI/ML model in the proctoring and evaluation pipeline
Generated 2026-07-06 · Scope: live model load + inference verification against real production data (not a statistical accuracy benchmark — see §6 Methodology) ·
Repository: Evaluation-Platform

1. Executive Summary

Model	Role	Test Result	Finding
YOLOv8n (<code>yolo8n.pt</code>)	Phone / object detection	PASS	Loaded and ran correctly on all 4 test frames. Correctly detected the person in every real candidate photo, correctly detected nothing in the blank frame, no false phone/object detections.
MediaPipe Face Detection + Face Mesh	Face presence / count / gaze basis	PASS	Correctly resolved face count = 1 on all 3 real candidate photos and 0 on the blank frame. Only passed once the exact pinned version (<code>mediapipe==0.10.14</code>) was installed — see finding below.
Claude Sonnet 5 (<code>claude-sonnet-5</code>)	Super-admin AI assistant	FAIL	API key authenticates correctly, but the request was rejected — Anthropic account credit balance is too low . This feature is non-functional in its current deployed state.
Claude Sonnet 4.6 (<code>claude-sonnet-4-6</code>)	Test generation / performance evaluation / trust reports	FAIL	Same billing failure. Every feature routed through <code>callLLM()</code> is currently non-functional.
GPT-4o-mini	Fallback LLM (unused unless <code>LLM_PROVIDER=openai</code>)	NOT TESTED	Inactive code path — <code>backend/.env</code> sets <code>LLM_PROVIDER="anthropic"</code> , so this default is never reached in the current configuration.

Critical finding: Both live Claude models returned `HTTP 400 invalid_request_error: "Your credit balance is too low to access the Anthropic API."` The API key itself is valid and correctly formatted (auth succeeded — this is a billing rejection, not an authentication failure). **Every LLM-dependent feature in this platform — AI test/question generation, candidate performance evaluation, trust report narratives, and the super-admin assistant — is currently unable to serve requests** until the Anthropic account is topped up.

Positive finding: Both computer-vision models used for live proctoring (face/person detection via MediaPipe, phone/object detection via YOLOv8n) loaded successfully and produced correct detections on real candidate webcam captures pulled from `backend/storage/objects/` . The proctoring pipeline's core detection logic is verified operational.

Version-pinning finding: Installing the *latest* published MediaPipe (0.10.35) instead of the version pinned in `python_cv_service/requirements.txt` (`mediapipe==0.10.14`) caused a hard failure — `module 'mediapipe' has no attribute 'solutions'` — because newer MediaPipe releases removed the legacy `solutions` API that `app.py` depends on. This is not currently breaking production (the pin is correct and respected in the requirements file), but it means **any future dependency upgrade of MediaPipe without an `app.py` rewrite will silently break every face-detection-based violation** (`face_not_detected`, `multiple_faces`, `looking_away`). Recommend pinning with an explicit upper bound or a code comment flagging this constraint.

2. Model Inventory & Test Matrix

Model	Type	Version / ID	Used By	Load Test	Inference Test
YOLOv8n	Object detection (CNN)	<code>yolo8n.pt</code> (Ultralytics 8.3.40)	<code>python_cv_service/app.py</code> — <code>phone_detected</code> , <code>unauthorized_object_detected</code>	✓ 63ms	✓ 4/4 frames, correct on all
MediaPipe Face Detection	Face bounding-box model	0.10.14, <code>model_selection=1</code>	<code>python_cv_service/app.py</code> — <code>face_not_detected</code> , <code>multiple_faces</code>	✓	✓ 4/4 frames, correct on all

MediaPipe Face Mesh	478-landmark face mesh	0.10.14	python_cv_service/app.py — cross-check for multiple_faces, basis for looking_away gaze calc	✓	✓ 4/4 frames, correct on all
Claude Sonnet 5	LLM (Anthropic)	claude-sonnet-5	superAdminAssistant.ts — admin chat assistant	✓ auth OK	✗ 400 — insufficient credit
Claude Sonnet 4.6	LLM (Anthropic)	claude-sonnet-4-6	llmService.ts default → testAgentService.ts , performanceEvaluationService.ts , trustReports.ts	✓ auth OK	✗ 400 — insufficient credit
GPT-4o-mini	LLM (OpenAI)	gpt-4o-mini	llmService.ts fallback default (inactive — provider set to anthropic)	N/A Not exercised — code path inactive	

3. Computer-Vision Model Smoke Test Detail

Test method: ran the exact detection code paths and thresholds from python_cv_service/app.py (MediaPipe dual-model cross-check logic, YOLO confidence 0.28 / IoU 0.45) against 3 real candidate webcam captures retrieved from live storage (backend/storage/objects/ , single person per frame — ground truth confirmed visually) plus 1 synthetic blank frame (ground truth: no person/face).

Test Frame	Ground Truth	MediaPipe Result	YOLO Result	Verdict
Candidate capture #1 (640×480)	1 face, 1 person, no phone	face_detection=1, face_mesh=1 → resolved=1 correct	person×1 (93.3%), no phone correct	PASS
Candidate capture #2 (1280×720)	1 face, 1 person, no phone	face_detection=1, face_mesh=1 → resolved=1 correct	person×1 (94.0%), no phone correct	PASS
Candidate violation-flagged capture (640×480)	1 face, 1 person, no phone	face_detection=1, face_mesh=1 → resolved=1 correct	person×1 (89.5%), no phone correct	PASS
Synthetic blank frame (640×480)	0 faces, 0 people	face_detection=0, face_mesh=0 → resolved=0 correct	0 detections correct	PASS

Inference latency observed

Model	Cold-start / first call	Warm calls (avg)
YOLOv8n	3,709–4,865 ms (first frame; includes CUDA/CPU backend init)	~200 ms
MediaPipe Face Detection + Mesh (combined)	~40 ms	18–38 ms

The YOLO cold-start cost (~4s on the very first frame of a session) is consistent with model graph compilation on first inference — this matches why python_cv_service/app.py loads the model once at process start rather than per-request. No cold-start penalty is paid per exam session under the current architecture.

Coverage gap — disclosed, not fabricated

No real multi-face or phone-visible test images were available in the local repository/storage to test the multiple_faces (count>1) and phone_detected =true paths end-to-end. Synthetic composites were deliberately not used for this claim, since neither YOLO nor MediaPipe reliably respond to drawn/synthetic shapes the way they do to real camera input — a "pass" against a synthetic multi-face image would not be a trustworthy accuracy signal. The single-face and no-face paths (the two most common real-world states) are verified above; the true-positive rate for multiple faces and phone detection remains unverified by this test and would require either consented multi-person/phone test recordings or the published Ultralytics/MediaPipe benchmark figures cited in §6.

4. LLM Smoke Test Detail (live Anthropic API calls)

Model	HTTP Status	Latency	Response
claude-sonnet-5	400	1,562 ms	invalid_request_error : "Your credit balance is too low to access the Anthropic API. Please go to Plans & Billing to upgrade or purchase credits."
claude-sonnet-4-6	400	862 ms	Same error — identical billing rejection.

Test method: minimal live `POST /v1/messages` calls (`max_tokens: 20`) using the real `LLM_API_KEY` configured in `backend/.env` , with the exact model ID strings referenced in the codebase. A 400 `invalid_request_error` for billing (rather than a 401 `authentication_error`) confirms the key format and permissions are valid — the account simply has no usable credit balance right now.

5. Proctoring Violation Matrix (current configuration)

Reference: `PROCTORING_VIOLATIONS_AUDIT.txt` , cross-verified against live code (`customAIViolations.ts` , `violationLabels.ts` , `proctoring.ts`).

Enabled — counts toward violation limit

Event	Detection Source	Threshold / Rule	Cooldown	Trust Report
tab_switch	Frontend visibilitychange; backend screenInfo	Tab hidden / not visible	None (5s UI dedupe)	No
window_blur	Frontend window blur event	Window loses OS focus	None	No
fullscreen_exit	Frontend fullscreenchange; backend screenInfo	Fullscreen not active	None	No
copy_paste_attempt	Frontend copy/paste intercept	Copy or paste attempted	None	No
devtools_open	Key-combo + window-size heuristic	F12/Ctrl+Shift+I/J, or outer-inner size delta >160px	None	No
camera_blocked	Frontend obstruction monitor; Python CV	Dark/uniform/disconnected feed >1200ms (FE) or >1.0s (Py)	Python 1.8s	No
face_not_detected	MediaPipe (Python); frontend AI fallback	face_count==0 for >1.5s (Py); 3 consecutive no-face frames (FE)	FE 12s; Py 1.8s	No
multiple_faces	MediaPipe dual-model cross-check (Python); frontend AI fallback	face_count>1 for >1.5s, both Face Detection AND Face Mesh must agree (min taken on disagreement)	FE 12s; Py 1.8s	Yes
looking_away	MediaPipe gaze (Python); frontend AI fallback	Off-screen gaze >1.5s, GAZE_LEFT_RIGHT_THRESHOLD=0.30	FE 12s; Py 1.8s	Yes
phone_detected	YOLOv8n (Python); frontend object rules	Confidence ≥0.28 and area ratio ≥0.00003 (Py); cell phone ≥35% area (FE)	FE 9000ms+12s; Py 1.8s×2	Yes
secondary_monitor_detected	Backend monitor-count transition	Transition from ≤1 to >1 monitor	FE 6000ms	No

Low priority — detected, does NOT count toward limit

Event	Rule	Trust Report
unauthorized_object_detected	laptop ≥45% or remote ≥45% area, or mapped object classes	Yes

Disabled (blocked by TEMP_DISABLE_AUDIO_PROCTORING / TEMP_DISABLED_EVENTS)

Event	Would-be Rule	Status
suspicious_audio	AUDIO_WARNING_LEVEL=42 / CRITICAL=58 / VOICE_MIN=16	Configured, blocked
voice_detected (+ 6 alias event IDs)	No active analyzer emit path	Configured, blocked

Defined only / info-only

Event	Status
eyes_closed_long, rapid_typing, unusual_mouse_movement, screen_share_stopped	No active detector implementation

external_monitor	Legacy alias → secondary_monitor_detected
camera_resumed, tab_switch_resume, window_focus_return	Info-only, never increments violation counter

6. Methodology & Limitations

- **What this report is:** a live smoke test confirming each model in the codebase actually loads with its pinned/referenced dependency version and produces correct output on real (CV) or real-auth (LLM) input, run against this specific repository on 2026-07-06.
- **What this report is not:** a statistical accuracy benchmark (precision/recall/F1 across a large labeled dataset). That would require a labeled corpus with known face counts, phone presence, and gaze direction across varied lighting/backgrounds/demographics — not available in this repository.
- **CV test data:** 3 real candidate webcam captures pulled from live storage (`backend/storage/objects/`), each manually confirmed by visual inspection to contain exactly one face/person and no phone, plus 1 synthetic blank frame for the negative case. Multi-face and phone-visible ground truth was not available locally.
- **LLM test:** minimal live API calls (20 max output tokens) against the real, currently-configured API key — not a mock or dry run.
- **Published third-party benchmarks (not independently re-verified here):** YOLOv8n reports ~37.3 mAP@0.5:0.95 on COCO val2017 (Ultralytics official benchmark); MediaPipe Face Detection (BlazeFace-based, `model_selection=1`) is optimized for faces up to ~5m from camera with reported real-time accuracy suitable for the <1m webcam range used here. These are Anthropic/Ultralytics/Google-published figures, not measurements taken by this test.
- **Environment:** Windows 11, Python 3.11.9, dependencies installed per `python_cv_service/requirements.txt` pins (opencv-python-headless substituted for opencv-python — headless build, functionally equivalent for this test since no GUI display calls were exercised).

Generated by an automated smoke test against the live repository at C:\Users\user\Desktop\New folder\git clone file\MAIN_BRANCH\Evaluation-Platform. Real candidate webcam images used for CV testing were processed locally only and are not reproduced in this report.